

A facet-exchange counterexample to the fixed-mass regular-simplex conjecture

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Abstract

We give an explicit family of positive four-cell mass vectors for which no translated regular-tetrahedral Gaussian partition maximizes the sum of squared first moments. This disproves the arbitrary-prescribed-mass statement of Conjecture 1.16 in Steven Heilman’s *Stable Gaussian Minimal Bubbles*, arXiv:1901.03934v1 (2019). The centering in its Problem 1.15 adds a constant under the prescribed-mass constraints. For a tetrahedral apex displaced toward one vertex, its two kinds of Gaussian facet measure are strictly unequal. The difference of the adjacent cell moments therefore has a nonzero tangential component. Exchanging two equal-measure small balls gives a genuine partition with exactly the same masses and strictly larger objective. We prove that every regular-tetrahedral partition with this mass vector is congruent to the displaced candidate, and that a measurable optimizer exists, so the refutation is not vacuous.

The equal-mass tetrahedral conjecture is not refuted or proved here. Related unequal-mass positive-noise results already exist; they are distinguished from the first-moment objective. No worldwide priority or external human-review assertion is made.

1. Exact conjecture and theorem

Let γ_d denote the standard Gaussian probability measure on $\mathbb{R}^d$, with density $\varphi_d(x) = (2\pi)^{-d/2} \exp(-\|x\|^2/2)$. For a measurable partition $\mathcal{C} = (C_1, C_2, C_3, C_4)$, define

$$p_i = \gamma_d(C_i), \quad m_i = \int_{C_i} x \, d\gamma_d(x), \quad \mathcal{E}(\mathcal{C}) = \sum_{i=1}^4 \|m_i\|^2. \quad (1)$$

Boundaries of the polyhedral cells below have Gaussian measure zero and may be assigned to the smallest label. We use Lebesgue integrals and Euclidean norms throughout.

Problem 1.15 in arXiv:1901.03934v1 fixes positive masses a_i summing to one, defines a common vector w from a translated regular-simplex partition, and writes $w^{(i)} = w/a_i$. Its objective, up to the common positive factor $\sqrt{\pi/2}$, is

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$$\sum_i \left\| \int_{C_i} (x - w/a_i) d\gamma_d(x) \right\|^2 = \sum_i \|m_i - w\|^2 = \mathcal{E}(\mathcal{C}) + 4\|w\|^2. \quad (2)$$

The last equality uses $\sum_i m_i = \int x d\gamma_d(x) = 0$. Thus the prescribed-mass maximizers are exactly those of (1), regardless of the fixed centering vector. Conjecture 1.16 states that for $m - 1 \leq d$ the maximizing sets in Problem 1.15 are simplicial cones over a regular simplex. Its $m = 4, d = 3$, arbitrary-positive-mass case is enough to refute the statement as written.

Use the regular-tetrahedral vectors

$$v_1 = (1, 1, 1), \quad v_2 = (1, -1, -1), \quad v_3 = (-1, 1, -1), \quad v_4 = (-1, -1, 1). \quad (3)$$

They sum to zero, have norm $\sqrt{3}$, and every pair is separated by $\ell = 2\sqrt{2}$. For each $t > 0$, put $w_t = tv_1$ and

$$C_i(t) = \{x : \langle v_i, x - w_t \rangle \geq \langle v_j, x - w_t \rangle \forall j\}, \quad p_i(t) = \gamma_3(C_i(t)). \quad (4)$$

The mass vector is specified by the fixed Gaussian integrals (4), not by numerical optimization. In particular $t = 1$ gives a single fixed counterexample instance with $w_1 = (1, 1, 1)$.

Theorem 1 (counterexample for every $t > 0$). For each mass vector $(p_1(t), p_2(t), p_3(t), p_4(t))$ in (4):

1. All entries are positive, $p_1(t) < 1/4$, and $p_2(t) = p_3(t) = p_4(t) = (1 - p_1(t))/3$.
2. There is a measurable four-cell partition with precisely these masses and strictly larger $\mathcal{E}$ than $\mathcal{C}(t)$.
3. Every regular-tetrahedral conical partition with precisely these masses, allowing any rotation and apex, has the same objective as $\mathcal{C}(t)$.
4. A maximizing measurable partition with these masses exists. None of the maximizing partitions can be regular-tetrahedral conical.

Consequently the arbitrary-mass Conjecture 1.16 just quoted is false. The same counterexample persists by cylindrical extension in every ambient dimension $d \geq 3$.

Corollary 1.1 (a full interval of mass vectors). For every $0 < p < 1/4$, regular-tetrahedral cones fail to maximize (1) at masses $(p, (1-p)/3, (1-p)/3, (1-p)/3)$. In particular, counterexamples occur arbitrarily close to the equal-mass vector. Indeed (5) and its null boundaries show that $p_1(t)$ is continuous, strictly decreases from $1/4$ to zero as t goes from zero to infinity, and hence has precisely that range. Continuity and the endpoint limit follow by dominated convergence. Theorem 1 supplies the strict improvement at every such t .

2. Exact geometry and the two facet measures

Write $\phi(u) = (2\pi)^{-1/2}e^{-u^2/2}$ and let Φ be its distribution function. Symmetries permuting v_2, v_3, v_4 and fixing v_1 are orthogonal and fix w_t . They give the three equal masses in Theorem 1. The first cell is

$$C_1(t) = \{(x, y, z) : x + y \geq 2t, x + z \geq 2t, y + z \geq 2t\}. \quad (5)$$

At $t = 0$ symmetry gives mass $1/4$ to every cell. For $t > 0$, (5) is strictly smaller up to a set of positive Gaussian measure. The set difference contains an open ball, for example around a point with all three coordinates strictly between 0 and t . Every cell contains a nonempty open set, so all four masses are positive.

For an interface between cells i, j , its Gaussian facet measure is

$$A_{ij}(t) = \int_{\Sigma_{ij}(t)} \varphi_3(x) d\mathcal{H}^2(x), \quad \Sigma_{ij}(t) = \overline{C}_i(t) \cap \overline{C}_j(t). \quad (6)$$

Triple intersections have surface measure zero. Symmetry leaves two values: $A = A_{12} = A_{13} = A_{14}$ and $B = A_{23} = A_{24} = A_{34}$. Both are strictly positive.

Lemma 2 (strict facet imbalance). For every $t > 0$,

$$A(t) = \phi(\sqrt{2}t) \mathbb{P}(X \geq t + |Y|/\sqrt{2}), \quad B(t) = \phi(0) \mathbb{P}(Z \leq \min(X/\sqrt{2}, 2t - X/\sqrt{2})), \quad (7)$$

where the Gaussian coordinates in each probability are independent standard normals. In particular $0 < A(t) < A(0) = B(0) < B(t)$.

Proof. The interface Σ_{12} lies in $y + z = 2t$. Use its orthonormal coordinates $X = x$, $Y = (y - z)/\sqrt{2}$, so $y = t + Y/\sqrt{2}$, $z = t - Y/\sqrt{2}$. The two remaining inequalities in (5) become $X \geq t + |Y|/\sqrt{2}$. Its normal coordinate is $\sqrt{2}t$, proving the first formula in (7).

The interface Σ_{23} lies in $x = y$. With orthonormal coordinates $X = (x + y)/\sqrt{2}$ and $Z = z$, the conditions that the common score of v_2, v_3 dominates v_1 and v_4 are respectively $Z \leq 2t - X/\sqrt{2}$ and $Z \leq X/\sqrt{2}$. Its normal coordinate is zero, proving the second formula.

The first probability strictly decreases as t grows from zero, and $\phi(\sqrt{2}t) < \phi(0)$. In the second formula the domain at $t = 0$ is $Z \leq -|X|/\sqrt{2}$, contained in the domain for $t > 0$. The inclusion is strict on an open set: take $0 < X < t/\sqrt{2}$ and Z close to zero. Gaussian densities are positive there. Finally $A(0) = B(0)$ by tetrahedral orthogonal symmetry. This proves every strict comparison. $\square$

3. First moments and a tangential obstruction

Lemma 3 (Gaussian flux formula). For the cells in (4),

$$m_i = \frac{1}{\ell} \sum_{j \neq i} A_{ij}(v_i - v_j). \quad (8)$$

Proof. The Gaussian identity $\nabla \varphi_3 = -x \varphi_3$ and the divergence theorem give the negative outward normal integral on each cell. On its ij facet the inward normal is $(v_i - v_j)/\ell$. To justify the unbounded cells, first intersect them with a radius- R ball. The artificial spherical boundary integral has magnitude at most a constant times $R^2 e^{-R^2/2}$, tending to zero. The moment and facet integrals converge absolutely. This gives (8). $\square$

Using (3) and the two facet values,

$$m_1 = \frac{4A}{\ell} v_1, \quad m_j = \frac{(A + 3B)v_j + (B - A)v_1}{\ell} \quad (j = 2, 3, 4). \quad (9)$$

Let $e = (1, 0, 0)$ and $n = (0, 1, 1)/\sqrt{2}$; the first is tangent to Σ_{12} and the second is its unit normal into C_1 . In these orthonormal directions the moment difference is

$$d = m_1 - m_2 = \sqrt{2}(A - B)e + (3A + B)n. \quad (10)$$

Thus $d \cdot e < 0$ by Lemma 2. A volume-constrained first variation would require the scalar score $d \cdot x$ to be constant along this facet. It is not constant. The next section does not rely on a formal variational argument: it constructs an actual exact-mass improvement.

4. An actual equal-measure ball exchange

Fix any $t > 0$ and its positive constants $A < B$. Define

$$\eta = \frac{B - A}{100(A + B)} \in (0, 1/100), \quad q = (t + 1, t, t) - \eta n, \quad r = (t + 2, t, t) + \eta n. \quad (11)$$

The open ball U centered at q of radius $\eta/2$ lies strictly inside $C_2(t)$, and the open ball V centered at r of the same radius lies strictly inside $C_1(t)$. Indeed their centers have distance η from the 12 plane on their respective sides. Distances from either center to the other relevant separating planes exceed $1/2$, by the explicit inequalities used in Section 2 and $\eta < 1/100$. Moving by $\eta/2$ cannot cross any of those planes. The two balls are disjoint.

Put $\delta = \frac{1}{2} \min(\gamma_3(U), \gamma_3(V)) > 0$. Gaussian measure of a concentric ball is continuous and strictly increasing in its radius from zero. There are therefore smaller concentric balls $U' \subset U$ and $V' \subset V$ with

$$\gamma_3(U') = \gamma_3(V') = \delta. \quad (12)$$

Transfer U' from cell 2 to cell 1 and V' from cell 1 to cell 2; leave cells 3 and 4 unchanged. This defines a measurable partition C' with exactly the same four masses. If

$$\Delta = \int_{U'} x d\gamma_3(x) - \int_{V'} x d\gamma_3(x), \quad (13)$$

its changed moments are $m_1 + \Delta, m_2 - \Delta$. We verify strict improvement rather than assume it from the tangent calculation.

By (10), $d \cdot (q - r) = \sqrt{2}(B - A) - 2\eta(3A + B)$ and $\|d\| \leq \sqrt{2}|A - B| + 3A + B < 5(A + B)$. For every $x \in U'$ and $z \in V'$, Cauchy–Schwarz consequently gives

$$\begin{aligned} d \cdot (x - z) &\geq \sqrt{2}(B - A) - 2\eta(3A + B) - \eta\|d\| \\ &> \left(\sqrt{2} - \frac{6}{100} - \frac{5}{100} \right) (B - A) > (B - A) > 0. \end{aligned} \quad (14)$$

Here $\sqrt{2} > 7/5$ is sufficient for the last inequality. Integrating (14) over the product of the two restricted Gaussian measures and using their common mass δ yields $d \cdot \Delta > \delta(B - A)$. Finally the exact norm identity gives

$$\mathcal{E}(\mathcal{C}') - \mathcal{E}(\mathcal{C}(t)) = 2d \cdot \Delta + 2\|\Delta\|^2 > 2\delta(B - A) > 0. \quad (15)$$

This proves part 2 of Theorem 1 with genuine Gaussian volumes and strict inequalities. No floating computation or solver output enters.

5. All regular-tetrahedral candidates with these masses

It is not enough to improve one chosen apex. We must rule out a different regular tetrahedron with the same mass vector.

Every regular-tetrahedral conical partition, after an orthogonal change of coordinates and common harmless scaling of scores, can be written

$$D_i(\lambda) = \{x : \langle v_i, x \rangle - \lambda_i \geq \langle v_j, x \rangle - \lambda_j \ \forall j\}, \quad \sum_i \lambda_i = 0. \quad (16)$$

Its apex is the unique w with $\lambda_i = \langle v_i, w \rangle$. The linear map $x \mapsto (\langle v_i, x \rangle)_i$ is a bijection from $\mathbb{R}^3$ onto the hyperplane $\sum_i s_i = 0$. The score vector has a strictly positive Gaussian density on that hyperplane and is invariant under every label permutation.

Lemma 4 (strict mass-price ordering). In (16),

$$\gamma_3(D_i(\lambda)) = \gamma_3(D_j(\lambda)) \iff \lambda_i = \lambda_j. \quad (17)$$

If $\lambda_i < \lambda_j$, the first mass is strictly larger.

Proof. Swapping labels i, j preserves the score distribution. Swap their prices to obtain λ' . Then the mass of cell i at λ' is the mass of cell j at λ . If $\lambda_i < \lambda_j$, changing from λ' to λ lowers the price of score i and raises that of score j , with other prices fixed. Every score vector won by i before is still won by i after. The inclusion is strict on an open subset of the score hyperplane: take s_i, s_j equal and sufficiently large, and all other scores sufficiently small, subject to zero sum. Before the change j wins, and after it i wins. Small perturbations retain both strict wins. The positive density gives strict mass increase. Equality of prices gives equality of masses by symmetry. $\square$

The three equal masses p_2, p_3, p_4 therefore force $\lambda_2 = \lambda_3 = \lambda_4$. Since their sum is zero, their prices have form $(3u, -u, -u, -u)$ and their apex is uv_1 . Also $p_1 < p_2$ forces $u > 0$. By (5), $p_1(u)$ is a strictly decreasing function of u : for $u_2 > u_1$, the cell inclusion is strict on an open ball centered at (b, b, b) with $u_1 < b < u_2$. Thus $p_1(u) = p_1(t)$ forces $u = t$. Orthogonal changes preserve all moments and norms. This proves part 3, including rotations and permutations of the three equal-mass labels.

6. Existence of a genuine maximizing partition

For completeness we give the existence argument, so part 4 and the refutation do not rely on an unproved assumption about attaining the supremum. Fix any positive masses p_i summing to one.

Consider the fractional partitions in the real Hilbert space $L^2(\gamma_3)^4$:

$$\mathcal{K} = \{(f_i) : 0 \leq f_i \leq 1, \sum_i f_i = 1 \text{ a.e., } \int f_i d\gamma_3 = p_i\}. \quad (18)$$

This is a nonempty bounded norm-closed convex set, and is weakly compact by Hilbert-space reflexivity. Each coordinate of $m_i(f) = \int x f_i(x) d\gamma_3(x)$ is a continuous linear functional because each Gaussian coordinate belongs to L^2 . Thus the objective, which depends on finitely many such coordinates, is weakly continuous and has a maximum M at some f^* . Let $m_i^* = m_i(f^*)$.

The linear functional $L(f) = \sum_i m_i^* \cdot m_i(f)$ satisfies $L(f) \leq \sqrt{M} \sqrt{\mathcal{E}(f)} \leq M$, with equality at f^* . Its maximizers form a nonempty weakly compact face of $\mathcal{K}$. The Krein–Milman theorem supplies an extreme point of this face, which is also extreme in $\mathcal{K}$.

Every extreme point of $\mathcal{K}$ is an indicator partition. Otherwise two components lie between ϵ and $1 - \epsilon$ on a common positive-measure region, for some $\epsilon > 0$ (use the finite number of component pairs and a countable union over positive rational ϵ). Gaussian measure is nonatomic. Split that region into two disjoint sets of equal positive measure and put $h = 1_{S_+} - 1_{S_-}$. Adding $\epsilon h/2$ to one component and subtracting it from the other, or reversing both changes, yields two distinct elements of $\mathcal{K}$ with the original as midpoint. This contradicts extremality.

Call the indicator extreme point g . Its $L(g) = M$ and Cauchy–Schwarz imply $\mathcal{E}(g) \geq M$ when $M > 0$. If $M = 0$, every member already has the maximum; a measurable partition with the prescribed masses is obtained by splitting an atomless probability space. In either case a genuine maximizing partition exists. The standard results used here are Hilbert-space reflexivity, weak compactness of bounded closed convex subsets of a reflexive space, and the Krein–Milman theorem [3]; none asserts Gaussian simplex optimality.

Part 2 gives a strictly better partition than the objective of every regular-tetrahedral candidate identified in part 3. The existing optimizer cannot be any of those candidates. This finishes Theorem 1. For $d > 3$, append independent unused Gaussian coordinates to the construction. Orthogonal reduction of any regular four-cell simplex to its three-dimensional span gives the same uniqueness argument. $\square$

7. Literature and exact scope of the refutation

The target is the **arbitrary-positive-prescribed-mass statement as written** in Heilman, *Stable Gaussian Minimal Bubbles*, arXiv:1901.03934v1, Section 1.4, Problem 1.15 and Conjecture 1.16, printed page 8 (13 January 2019) [1]. Its equal-mass specialization is different and is untouched by the present counterexample.

Heilman–Mossel–Neeman, *Standard Simplices and Pluralities are Not the Most Noise Stable*, Israel Journal of Mathematics 213 (2016), 33–53, arXiv:1403.0885v3, Theorem 2.6 [2], already proves a nonoptimality statement for unequal masses and every positive noise parameter $0 < \rho < 1$. That theorem statement does not itself assert strict improvement of the first-Hermite coefficient: a strictly positive function for $\rho > 0$ may have zero derivative at zero. Here strict first-moment improvement is proved directly by (15). No novelty claim is based merely on taking a limit of the known noisy result.

The finite literature comparison is in `PRIOR_WORK.md`; the archived primary-source hashes are in `SOURCES.json`. We do not claim this obstruction has never been observed. The proof uses Gaussian integrals and elementary inequalities, with no computational theorem input. The partial Lean check verifies algebraic identities and the strict exchange-margin reduction, not the full Gaussian geometry or existence theorem. All mathematical claims above have explicit written proofs.

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References

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- [3] W. Rudin, *Functional Analysis*, second edition, McGraw–Hill, 1991. The standard weak-compactness and Krein–Milman results are used for optimizer existence.